

Damietta University
Faculty of Computers and
artificial Intelligence



Mid Term Exam 2024-2025

15 Marks
Course: Data
Course Cod
Date: Wed

Student name: **Student ID:**

Answer the following:

- 1- Discuss Data Pre-processing.
- 2- Give 3 examples of messy data, explain how to manipulate it?
- 3- Complete the following table:

data	z-score normalized data
3	
8	
11	
15	
17	
20	
30	
130	

2) Explain the concept of data preprocessing. What are the techniques used for preprocessing of data?

- **Data preprocessing** transforms raw/inconsistent data into a usable format by addressing issues like missing values, noise, and redundancies.
- **Techniques:**
 - a. **Data Cleaning** : Remove duplicates/irrelevant data – fix structural errors – handle missing data – smooth noise
 - b. **Data Integration** : Combines data from multiple sources (ETL, data virtualization, warehousing)
 - c. **Data Transformation** : Standardizes data into main form
 - d. **Data Reduction** : Dimensionality reduction -- Numerosity reduction -- Data compression

5. Define Messy data, and give examples

- ◆ **Missy Data** : is data that is not in usable form or it is impossible to obtain clearly information from it
- ◆ **It needs lot of preparing/preprocessing to be usable**
Ex : Duplicate records with inconsistent spellings - column mixing two different data – missing data

Q: Give examples of messy data and how to manipulate it ? “Midterm”

Example	How to Manipulate it
Missing values (NaN or blanks)	Fill or drop missing data “df.dropna()”
Duplicate rows	Remove duplicates “df.drop_duplicates()”
Mixed data types in a column	Convert to a single data type
Combined data in one column	Split into multiple columns
Irregular date formats	Convert to datetime

Mean = 29.25

data	(X-M)	(X-M) ²	Z = $\frac{X-M}{\sigma}$
3	(3-29.25) = -26.25	686.44	Z = $\frac{-26.25}{38.8} = -0.68$
8	(8-29.25) = -21.25	451.56	Z = $\frac{-21.25}{38.8} = -0.54$
11	(11-29.25) = -18.25	333.0	Z = $\frac{-18.25}{38.8} = -0.47$
15	(15-29.25) = -14.25	203.06	Z = $\frac{-14.25}{38.8} = -0.36$
17	(17-29.25) = -12.25	148.8	$\frac{-12.25}{38.8} = -0.31$
20	(20-29.25) = -9.25	85.56	$\frac{-9.25}{38.8} = -0.23$
30	(30-29.25) = 0.75	0.56	$\frac{0.75}{38.8} = 0.01$
130	(130-29.25) = 100.75	10150.5	$\frac{100.75}{38.8} = 2.5$
	$\Sigma = 12058.8$		

Mean = $\frac{3+8+11+15+17+20+30+130}{8} = 29.25$

$\sigma = \sqrt{\frac{\Sigma(X-M)^2}{n}} = \sqrt{\frac{12058.8}{8}} = 38.8$